

Articulatory synthesis: nasal sounds and male and female voices

D. G. Childers and C. Ding

Department of Electrical Engineering, University of Florida, 405 CSE, Gainesville, FL 32611, U.S.A.

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We describe the implementation of an articulatory speech synthesizer using a discrete circuit model that converts the acoustic equations into linear algebraic equations. The algorithm is fast, flexible and stable. Three experiments were conducted with the synthesizer to determine: (1) the effect of the sinuses on the quality of nasal sounds; (2) the effect of the velopharyngeal port opening on nasal sounds; (3) the effect of fundamental frequency, vocal tract area, vocal fold parameters and excitation on the synthesis of female voices.

1. Introduction

Articulatory speech synthesis has appealed to researchers for years as a means for producing high quality synthetic speech. This appeal is based on the premise that replication of the mechanisms of human speech production should yield natural sounding speech, and should provide a mechanism for studying the relationships between control parameters, speech production and perception (Rubin, Baer & Mermelstein, 1981; Rothenberg, 1981; Koizumi, Taniguchi & Hiromitsu, 1985). Articulatory parameters are more easily interpolated than linear prediction (LP) or formant synthesizer parameters because the interpolated articulatory control parameters are physically realizable and consequently do not result in unnatural speech (Sondhi & Schroeter, 1987). The potential for bit rate coding below 4800 bits/s using the articulatory synthesizer has been noted (Flanagan, Ishizaka & Shipley, 1975, 1980; Schroeter, Larar & Sondhi, 1987).

Articulatory synthesis might be classified into three approaches. One represents the vocal tract as a concatenation of acoustic tubes with frequency dependent losses (Kelly & Lochbaum, 1962; Maeda, 1977; Rubin *et al.*, 1981; Strube, 1982). This approach offers computational savings and may take advantage of parallel processing techniques. However, it has at least four weaknesses. First, the frequency dependent propagation losses and radiation are modeled as resistive. Second, source–tract interaction is not included because the kinetic resistance of the glottis and the glottal volume-velocity are not modeled. Third, the vocal tract length cannot be varied easily since the length of each tube segment is fixed and related to the sampling interval. Fourth, the vocal tract is modeled with hard walls.

The second approach models the acoustic properties of the glottis and vocal tract with a set of ordinary differential equations (Flanagan & Landgraf, 1968; Ishizaka &

Flanagan, 1972; Flanagan *et al.*, 1975, 1980; Rothenberg, 1981; Koizumi *et al.*, 1985). This model has many advantages and attempts to include some physiologically relevant features; however, it is computationally expensive and the frequency dependent losses and radiation loads cannot be simulated with accuracy. Maeda (1982a) simplified Flanagan's model by omitting the self oscillating vocal fold portion of the model and substituting glottal area control parameters, by omitting noise sources within the vocal tract, and by neglecting losses due to viscous and heat conduction. These three simplifications made Maeda's algorithms much faster than in earlier models. Other simplifications to Flanagan's model were introduced by Bocchieri (1983) and Bocchieri & Childers (1984), who reduced the number of noise sources and modeled the vocal tract variation by drawing a sequence of tract images on a graphics terminal. This sequence of images attempted to replicate the human articulatory gestures. These authors also suggested improved algorithm implementation procedures.

The third approach to articulatory speech synthesis is a hybrid time-frequency domain method (Allen & Strong, 1985; Sondhi & Schroeter, 1987). The action of the vocal folds is modeled in the time domain while the vocal tract variations are modeled in the frequency domain, thereby providing certain computational and flexible realization advantages. However, this method does not yet synthesize high quality fast transition sounds, e.g., stops. In addition the synthesizer output provides only a speech waveform; it does not provide pressure or volume-velocity calculations.

For this study we selected the second approach since it offered the ability to simulate the dynamic properties of the vocal system (source and tract) more accurately than the other approaches and provided more acoustic information, in that we could simulate source-tract interaction and we could potentially synthesize all phonemes adequately. The research had three goals: first, to design a fast, flexible and stable algorithm for solving the numerous differential equations; second, to model and investigate factors relevant for synthesizing nasalized vowels and consonants; and third, to investigate the factors involved in modeling and synthesizing female voices.

2. The articulatory model

2.1. Model of the glottal source

Our model simulates the acoustic properties of the vocal folds, while the movement of the vocal folds is specified by the user via a glottal area function (Titze, 1984; Cranen, 1987). This simplification is based on two factors: first, the glottal area is almost independent of the vocal tract shape, while the vocal tract configuration may substantially alter the glottal volume-velocity waveform (Ishizaka & Flanagan, 1972); and second, vertical phasing of the vocal folds during vibration may make the projected glottal area relatively insensitive to tract loading (Allen & Strong, 1985).

For simulating aspiration, a noise pressure source P_{ng} was placed at the interface between the expansion of the vocal folds and the first section of the vocal tract. The value of P_{ng} is proportional to the difference of the square of the Reynolds number, Re^2 , and a critical Reynolds number, Rec^2 . Thus,

$$P_{ng} = \begin{cases} G \times \text{RANDOM} \times (Re^2 - Rec^2), & Re > Rec \\ 0, & Re \leq Rec \end{cases} \quad (1)$$

where G is an empirically determined gain, $RANDOM$ is a random number uniformly distributed between -0.5 and 0.5 , and Rec is 2700 (Sondhi & Schroeter, 1987).

The advantages of our glottal source model are four: first, it allows a reduction in computation by eliminating the nonlinear mechanical properties of the vocal folds; second, the user may specify an accurate pitch contour; third, it uses a glottal area function (obtained by measurement or simulation) as a control parameter; and, fourth, it has the capability to produce breathy voices.

2.2. Model of the vocal and nasal tracts

A section of the vocal tract may be represented by a concatenation of a finite number of transmission line elements whose elemental structure is given in the top panel in Fig. 1. The various pairs of nasal sinuses (3 to 4) are simulated with a Helmholtz resonator (Lindqvist & Sundberg, 1972; Maeda, 1982*b*). Excitations

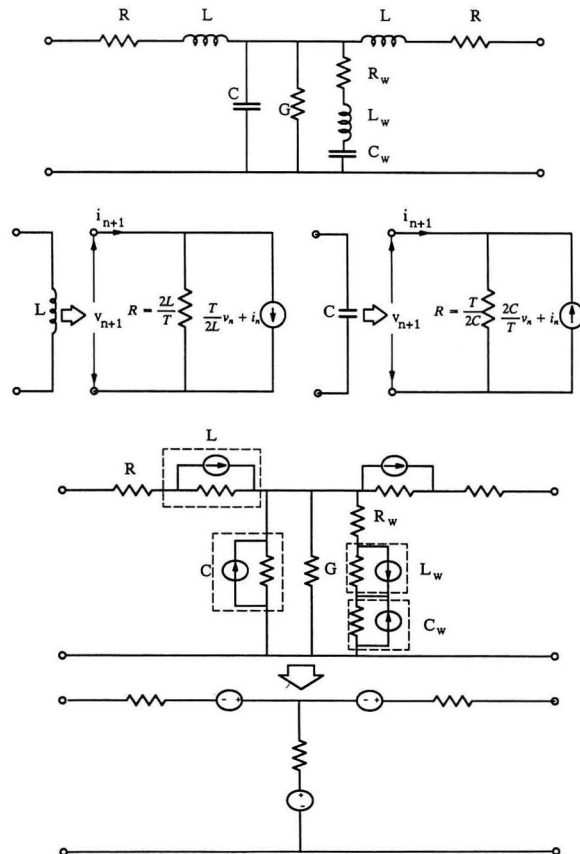


Figure 1. A typical network representation of an element in the analog transmission line model of the vocal tract. Discrete circuit model associated with trapezoidal algorithm for a linear inductor and capacitor and the simplified equivalent circuits of an element of the vocal tract using discrete circuit models and circuit analysis techniques.

caused by vocal tract constrictions are modeled as a noise source, U_n , in parallel with the vocal tract network. The intensity of the noise current is proportional to the square of the Reynolds number in excess of a specified critical value. The current source is placed one section downstream of the narrowest construction. The speech pressure is calculated as the differential of the sum of volume-velocities at the lips and nostrils, using a conventional radiation model (Flanagan, 1972). The effect of different phase relationships between the output of the nostrils and the lips is discussed later.

2.3. Numerical method

The above model is implemented as a set of ordinary differential equations which may be categorized as either non-stiff or stiff equations (Gear, 1971; Gupta, Sacks-Davis, & Tischer, 1985). Stiff equations are characterized by a Jacobian with disparate eigenvalues and with some eigenvalues possibly having large negative real parts (Gear, 1971; Shampine & Gear, 1979). We identified the acoustic equations as falling into the stiff category and consequently we were able to devise a very efficient method of solution. Furthermore, the solution of the stiff equations was made more stable by using the trapezoidal algorithm for solving $y' = f(y)$, namely

$$y_{n+1} = y_n + \frac{T}{2} [f(y_n) + f(y_{n+1})]. \quad (2)$$

The discrete circuit model may be used to simplify the acoustic equations (Chua & Lin, 1975). Figure 1 shows how the acoustic equations are transformed into a set of

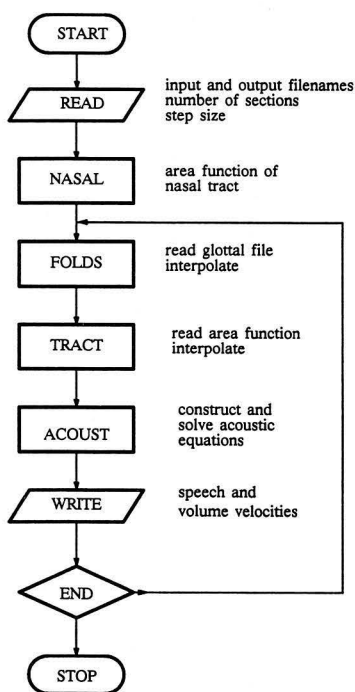


Figure 2. Block diagram of the articulatory synthesizer program, ART_SYN.

linear algebraic equations with a sparse coefficient matrix using circuit analysis techniques.

2.4. Realization

The computer program implementation lets a user specify the input and output filenames, the number of sections in the acoustic transmission line, and the step size of the trapezoidal method. The two input files comprise the area function file and the glottal source file, while the two output files comprise a speech file and a file that contains volume-velocities and pressure waveforms at the glottis, mouth and nostrils. The user also specifies the nasal tract area function. A block diagram of the program is shown in Fig. 2. A comparison of our synthesizer with three other similar synthesizers appears in Table I (Flanagan *et al.*, 1975; Maeda, 1982a; and Bocchieri & Childers, 1984).

Our implementation is simple computationally: it allows the user the freedom to specify an accurate pitch contour, it is able to simulate source-tract interaction, it is stable, efficient, flexible and easily used. A typical experiment to synthesize a sentence takes only a few minutes to set up the necessary files and about 4 min to produce the synthesized sentence on a SUN 3 Microsystems workstation. One advantage of the model is that the vocal tract length can be dynamically varied during the synthesis of a sentence.

TABLE I. Comparison of several synthesizers.

	Flanagan	Maeda	MMIRC-1*	MMIRC-2*
Model of vocal folds	Self oscillating two-mass model	A slit	Self oscillating two-mass model	Passive one-mass model
Noise source at the glottis	No	No	Yes	Yes
Source parameters	A_{go}, Q, P_s	A_p, F_0, P_s	A_{go}, Q, P_s	$A_{go}, A_{gamp}, Q_0, Q_s, F_0, P_s$
Noise source in the tract	Every section	No	At constriction	At constriction
Viscous and thermal losses	Yes	No	Yes	Yes
Tract parameters	$A(x)$	$A(x)$	$A(x)$ or position of articulators	$A(x)$ or position of articulators
Method of simplification	From equations to equations	From equations to equations	From equations to equations	Circuit models
Numerical method	Backward Euler	Integration by mid-point rule	Runge-Kutta	Backward trapezoidal
Method of change of model parameters	Re-compile source code	Re-compile source code	Re-compile source code	Through parameter files

Source parameters: A_{go} , minimum glottal area; Q , cord tension; P_s , subglottal pressure; A_p , peak glottal area; F_0 , fundamental frequency of voicing; A_{gamp} , same as A_p ; Q_0 , open quotient; Q_s , speed quotient; $A(x)$, vocal tract cross-sectional area function.

* MMIRC: Mind Machine interaction Research Center. MMIRC-1: Bocchieri & Childers (1984); MMIRC-2: this paper.

3. Experimental results

3.1. Nasal tract and velum

The degree of closure of the velopharyngeal port varies according to the phonetic context, e.g., from the open position characteristic of nasals, to the intermediate position characteristic of low vowels, to the more nearly closed position characteristic of high vowels, to the closed position typical of obstruent consonants. Bjork's (1961) tomographic and cineradiographic study concluded that the velopharyngeal port area is a linear function of the port's sagittal minor axis, and that the constant of proportionality is 10 mm. Hence, the anatomical structure of the port is nearly rectangular. Several investigators (Bjork, 1961; Warren, 1967; Isshiki, Honjow, & Morimoto, 1968) are of the opinion that the linguistically useful size of the velopharyngeal port lies within the range from zero to slightly more than 1 cm^2 . A general rule is that when the velum comes within 2 mm of the pharynx (producing an open area of about 20 mm^2) there is no apparent nasality, while a wider opening (5 to 50 mm^2 in area) produces nasal resonances, with the resulting speech being definitely perceived as nasal (Borden & Harris, 1984).

3.2. Synthesis experiments

In the following two experiments, speech and electroglottographic (EGG) data for a male speaker of American English were digitized for the sentence *Ben went mining*. The signals were collected in an IAC sound booth, digitized at a sampling frequency of 10 kHz with 16 bit accuracy. For details of data collection refer to Childers & Wu (1990).

The four features extracted from the speech and the EGG signals were: pitch contour; intensity contour; glottal open quotient; and formant frequencies as a function of time. The articulatory information was obtained heuristically from phonetic considerations and from X-ray data available in the literature (Fant, 1960; Hecker, 1962; Levinson & Schmidt, 1983), while the phonation information was derived from the pitch contour, the intensity contour and the glottal open quotient.

Based on the features extracted from the speech signal and the area functions reported in the literature (Fant, 1960), the synthetic sample was produced using a trial-and-error method. After obtaining satisfactory synthetic speech, listening tests were conducted to investigate the effects of the maxillary sinuses and the opening of the velopharyngeal port on the quality of the nasal sounds. The judges for the listening tests were three professors familiar with both vocal disorders and synthetic speech.

3.3. Experiment I

Experiment I was conducted to study the effect of the sinuses on the quality of nasal sounds. Figure 3 shows the area functions of the nasal tract and the maxillary sinuses used in the experiment. The sinuses are connected to the nasal tract at 7 cm from the velopharyngeal port via a 0.1 cm^2 opening, similar to Maeda (1982b). This connection was simulated with a single RLC circuit. The sentence *Ben went mining* was synthesized with and without the maxillary sinuses and with five different

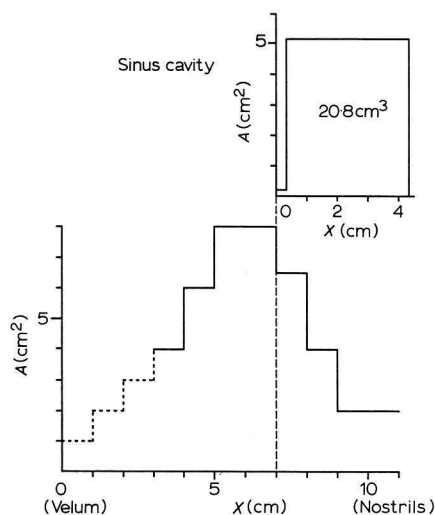


Figure 3. The area function of the nasal tract.

velopharyngeal openings (0.1, 0.2, 0.4, 0.6 and 1.4 cm²) The listening tests were similar to those described in Childers & Wu (1990) and were intended to determine which conditions resulted in a more natural sentence with respect to nasality. Figure 4 shows the waveforms and the spectra of a nasal consonant /m/ synthesized with and without the sinuses. The antiformant near 1800 Hz was more prominent in the spectrum for the nasal synthesized by simulating the effect of the maxillary sinuses. However, the waveforms of the nasalized vowels were nearly identical. The listening tests indicated that the simulation of the maxillary sinuses did not greatly affect the quality of the nasal sounds. This result differs from Maeda's (1982*b*) result. He reported that when an acoustic tube having the appropriate area function with no

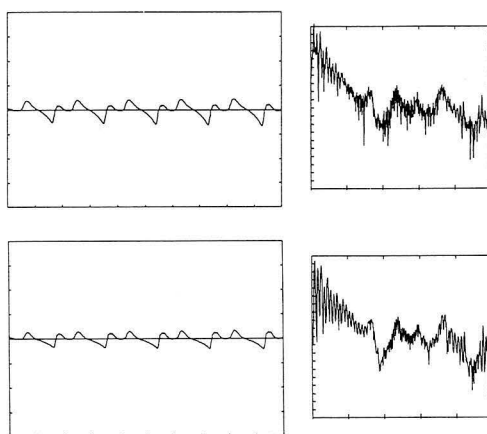


Figure 4. The acoustic waveforms (left) and corresponding spectra (right) for the nasal sound /m/ synthesized without (top) and with (bottom) maxillary sinuses. The duration of the acoustic signal is 60 ms and the spectra ranges from -10 to 90 dB with a frequency scale of 0 to 5000 Hz.

side branching cavities (sinuses) was used to simulate nasals, the low frequency nasal formant below the first formant of the low vowels (as observed in natural speech) could not be simulated. Furthermore, the vowel quality was modified with the synthetic nasalized mid and low vowels not sounding nasalized.

There are two reasons why the maxillary sinuses may not affect the quality of synthesis of nasalized vowels. First, the input impedance of the nasal tract at the velopharyngeal port changes very little when the sinuses are added, since the nasal tract already introduces considerable damping. The maxillary sinus pair is near the nostrils and thereby introduces little additional damping. Second, the volume-velocity at the nostrils is much smaller than the volume-velocity at the lips for nasalized vowels. The sound pressure is the differential of the sum of these two volume-velocities. Although the volume-velocity at the nostrils does change, the sound of nasalized vowels changes only slightly, since the volume-velocity at the nostrils is a small part of the sum.

3.4. *Experiment II*

The purpose of this experiment was to correlate the nasality of synthesized nasals to the opening of the velopharyngeal port. The stimuli of this experiment were the same as in Experiment I, including the simulation of the maxillary sinuses. The listening tests were similar to those for Experiment I.

The results of the perceptual listening test indicated that the perception of nasality was related to the opening of the velopharyngeal port. In general the perception of nasality increased with an increase in the opening of the velopharyngeal port. The spectra of the nasal consonant /m/ (see Fig. 4) showed that the first zero (which appeared in conjunction with an additional formant) occurred at about 1.8 kHz, and also that the first, second, and third formants occurred approximately at 250, 900 and 2400 Hz, respectively. (Small differences in the spectra in Fig. 4 below 500 Hz are not consistent and should be ignored.) These values were in agreement with Fujimura's predictions (Fujimura, 1961). With an increase of the opening of the velopharyngeal port, the valleys in the spectra became more pronounced. Figure 5 illustrates the waveforms and the spectra of a nasalized vowel for different openings of the velopharyngeal port. Our results were generally in agreement with Bjork (1961), Warren (1967) and Isshiki, *et al.* (1968), indicating that our articulatory speech synthesizer could simulate the effects of the nasal tract. Although the differences in the spectra in Fig. 5 due to the variations in the velopharyngeal openings are small, the differences are acoustically perceptible, suggesting that to synthesize nasal sounds the velopharyngeal opening should be 0.4–0.6 cm².

3.5. *Properties of female vocal system*

The different formant patterns between the sexes are due mainly to the size of the vocal tract. According to Fant (1973), the ratio of the total length of the female vocal tract to that of a male vocal tract is about 0.87.

Additional gender differences may be due to the strength of the fundamental component and the general tilt of the harmonic spectrum caused by the source (Rosenberg, 1971; Holmes, 1973; Kitzing & Sonerson, 1974; Childers, Wu, & Hicks, 1987; Klatt, 1987; Childers & Wu, 1990; Klatt & Klatt, 1990). The source

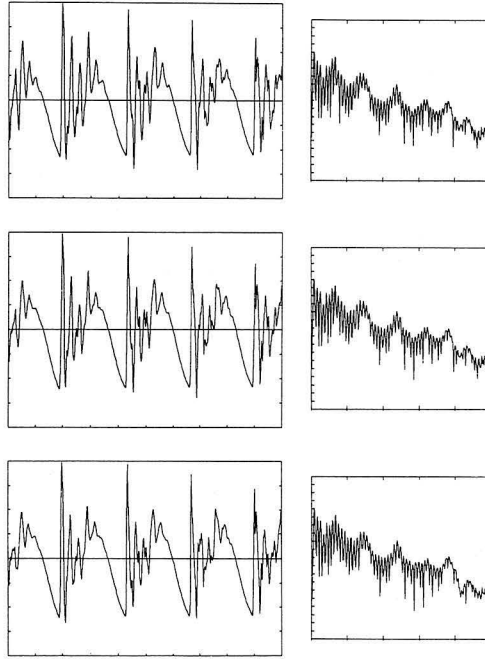


Figure 5. Speech waveforms (left) and spectra (right) of a nasalized vowel /a/. The three rows are for different areas of velopharyngeal opening: 0.1 cm² (top), 0.4 cm² (middle) and 1.4 cm² (bottom). The duration of the speech signal is 50 ms. The spectra range is from 20 to 120 dB with a frequency scale of 0 to 5000 Hz.

characteristics should include features to simulate breathiness, a factor often associated with female voices (Lee & Childers, 1989; Childers & Wu, 1990; Klatt & Klatt, 1990).

3.6. Experiment III

Voice conversion was selected as the method for synthesizing female voices (Childers, Wu, Hicks & Yegnanarayana, 1989). Sentences were first synthesized to replicate a male voice; then a female voice was synthesized by systematically varying the control parameters following the procedures in Childers *et al.* (1989). Listening tests similar to those in Childers & Wu (1990) were conducted to evaluate the perceptual correlations of the control parameters to the synthesis of female voices. Two sentences, *Good-bye Bob* and *We were away a year ago*, were synthesized.

Four groups of control parameters were investigated in this experiment: the fundamental frequency, the vocal tract area function, the vocal fold parameters (length, thickness and the glottal area waveform) and the generation of noise at the glottis. After replicating the male voice, the control parameters were systematically varied to generate female voices. No attempt was made to match a specific target female voice. Following the above procedure of varying each parameter individually and all possible combinations, we synthesized 15 “female” voices from the one male

voice. The fundamental frequency was scaled by 1.9. The length of vocal tract area was scaled as follows: the pharyngeal section was scaled by 0.8 and the oral section by 0.9. The area of the pharyngeal section was scaled by 0.64 and the oral section by 0.81. The length and thickness of the vocal folds was scaled by 0.8. In addition we used one excitation waveform for the male voice and a more symmetrical one for the female voice. Finally, we simulated breathiness by varying the amount of glottal noise (Lee & Childers, 1989). We found that the simulation of breathy voices could be controlled by a noise generator at the glottis, which produced a slightly breathy voice. If a sinewave-like glottal area function was used, then the perception of the degree of breathiness increased. Using both a sinewave-like glottal area function and a noise generator resulted in the perception of the highest degree of breathiness. The noise generator is controlled by the Reynolds number and the glottal noise is amplitude modulated. Lee & Childers (1989) have reported that amplitude modulation has a significant effect on achieving perceptual naturalness of vowels for breathiness. They also reported that high-pass filtering of the noise source had little effect on perceived naturalness, as did the location of the noise production within the pitch period. Their experiments were conducted using only synthesized vowels. Our experiments with synthesized sentences indicated that the noise was most effective when generated at the peak of the glottal volume-velocity, which agrees with Stevens (1971). While more studies are needed to better explain these issues, it appears that the synthesis of coarticulation effect may be affected by the temporal location of the noise source within the glottal cycle. We also varied the average value of the pitch contour, the open quotient, Q_o , the speed quotient, Q_s , the minimum glottal area A_{go} , and, as mentioned above, the scaling of the pharyngeal and oral tracts and the vocal tract.

4. Summary

In summary, our experiments may clarify some previous controversy concerning various parameters discussed in Coleman (1976) and Lass, Hughes, Bowyer, Waters & Bourne (1976). We determined that pitch contour is the most important parameter for synthesizing female voices, followed by vocal tract length and glottal excitation factors. The data from the experiments suggest that an average pitch of 190 Hz is preferred. In addition, the vocal tract area function, especially the vocal tract length, is a key control parameter for producing female voices. A vocal tract length in the range from 14.2 cm to 15.3 cm appears proper for female voices. The waveshape of the glottal area function is another key control parameter for synthesizing female voice characteristics. An open quotient larger than 0.8 and a speed quotient near 1 are preferred. To generate a spectral tilt of more than 12 dB per octave, which is important for female voices, the glottal area function should have a "rounded corner" at glottal closure. A turbulent noise source at the glottis is a key factor for producing a breathy voice, which was preferred for female voices.

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